

Hyperparameter Note-Taking Framework

Cole Drumheller

Wednesday 27th August, 2025 - 00:58

1 Summary

The material covered a few different topics, the difference between parameters and hyper-parameters, different methods of tuning hyper-parameters, and optimizing hyper-parameters using Bayesian optimization. First, we get into the differences between parameters and hyper-parameters. In simple terms, parameters are the values that are learned during machine learning. In contrast, hyper-parameters are manually specified before training, they heavily impact the model's performance, and they ultimately help to determine the final values of the parameters.

The videos also introduced several methods of figuring out the best values to set for the hyper-parameters in order to get the best results from the models. The first method proposed was to simply experiment with different values by hand and see what gives the best results, also known as an Empirical Process (Parameters vs Hyperparameters (C1W4L07), 3:05-5:00). Another method that was discussed was grid search. When doing grid search you choose X amount of values for each different hyper-parameter, then you create and train/test models on all the possible combinations of those hyper-parameters (The Ultimate Guide to Hyperparameter Tuning — Grid Search vs. Randomized Search, 1:30). You then select the model that performs the best. The last tuning method discussed was Randomized Search. This method is similar to grid search, but instead of the user defining discrete values they give a range for each hyper-parameter along with the number of models they want created. The method then randomly samples from the given ranges for the defined number of models (The Ultimate Guide to Hyperparameter Tuning — Grid Search vs. Randomized Search, 2:35). Comparing grid search and randomized search, grid search is typically more effective when you have a small number of hyper-parameters and a narrow range, while randomized search can be more effective when you have many hyper-parameters with wider ranges (The Ultimate Guide to Hyperparameter Tuning — Grid Search vs. Randomized Search, 3:17).

The final topic that was discussed was Bayesian Optimization. This introduced a way to optimize the hyper-parameters for a model. It starts by evaluating initial points and building a surrogate model using a gaussian process to identify areas of certainty/uncertainty. Bayesian Optimization then uses

acquisition functions to balance the exploration and exploitation tradeoff. The process focuses more on exploration to start, then as the uncertainty decreases and it gets closer to the optimal value, it begins to favor exploitation. The process then evaluates its new points, updates its model and repeats the process until it finds the optimal values (Bayesian Optimization, 7:20). This method of updating after each increment is called active learning and it's what sets Bayesian Optimization apart from your standard grid search or randomized search.

2 Attention Grabbers

1. I have never heard someone explain hyperparameters the way the guy in the second video did right off the bat. He said, "Parameters that control W and b (the main parameters) so we call these things hyperparameters" (Parameters vs Hyperparameters (C1W4L07), 1:37).
2. I thought the Emperical process (Parameters vs Hyperparameters (C1W4L07), 3:05-5:00) that the guy in the second video talked about sounded interesting. However, having worked with some of these models before, I can't imagine trying to manually find the best hyperparameters for a more complex model.
3. The third video talked about grid search (The Ultimate Guide to Hyperparameter Tuning — Grid Search vs. Randomized Search, 1:30) which is something that I've done a lot of, so that got my attention and made a lot of sense. They also talked about how long it can take, which I have experienced first hand.
4. Also in the third video was the randomized search (The Ultimate Guide to Hyperparameter Tuning — Grid Search vs. Randomized Search, 2:35). I found this topic interesting because I had never heard of this before, which surprised me because I've taken quite a few machine learning type classes.
5. I really liked the visuals used throughout the 4th video (Bayesian Optimization). I thought it did a really good job of not only saying how each method worked, but the visualization was really good and made it make more sense.

3 Questions

1. What scenarios would a Bayesian optimization be worse than Grid Search or Random Search?
2. The first video talks briefly about SVCs, and it says that C , γ , and kernel are regular parameters not hyperparameters (0:54). Why are these not hyperparameters since they are manually specified, not learned?

3. When doing Bayesian optimization, what are some ways to make sure the process doesn't just sit on a local maximum?
4. What are the situations to use the different acquisition functions talked about in the Bayesian Optimization video (3:38)?
5. The Bayesian optimization only talked about optimizing accuracy, sometimes high training accuracy can mean that the model is just overfit to the data, how can you avoid this?